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SignSpotter: scene text-informed cross-modal semantic consistency verification for street view images and POI data

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ABSTRACT

Ensuring semantic consistency between street view images and Points of Interest (POI) is critical for reliable geographic entity validation in complex urban scenes. To address challenges such as visual occlusion, scale variation, and ambiguous scene text, this paper proposes SignSpotter, a novel multi-scale vision-language framework for entity-level consistency verification. First, a scene text-informed visual pre-training strategy is introduced to sensitize the Vision Transformer to fine-grained textual regions. Second, a resolution-adaptive asymmetric dual-tower architecture is employed to jointly model multi-scale visual features and POI textual embeddings, effectively mitigating scale disparities. Furthermore, a cross-modal semantic reasoning module establishes precise visual-textual associations via bidirectional attention for explicit consistency inference. Experiments on a large-scale dataset demonstrate that SignSpotter achieves an F1-score of 90.96% on the source city dataset. Cross-city and cross-platform transfer tests further yield stable F1-scores exceeding 86%. These results, together with occlusion analyses, confirm the robustness of SignSpotter and highlight its potential for automated large-scale geographic data quality control.

1. Introduction

With the proliferation of mobile internet technologies and LBS, User-Generated Geographic Content (UGGC) has emerged as a pivotal data source for enriching map databases and facilitating the dynamic updating of Points of Interest (POI) (Goodchild, 2007; Yan et al., 2020). Among these, user-uploaded street view imagery serves as a direct terrestrial remote sensing interface to the physical environment, offering critical visual evidence for verifying the authenticity of POI attributes (Tang and Painho, 2023; Chen et al., 2025). However, the unstructured nature and inherent noise of UGGC generation often induce semantic discrepancies between street view images and their corresponding POI records. Such inconsistencies fundamentally compromise the reliability of map services and the credibility of downstream applications (Senaratne et al., 2017; Li et al., 2022). Consequently, developing efficient and automated mechanisms for semantic consistency verification between street view imagery and POI data has become a pressing challenge in intelligent urban sensing and geospatial data quality control. In this paper, the target task is formulated as an entity level binary matching problem. Specifically, given an input real world street view image and a corresponding target POI textual attribute, the model is required to explicitly output a binary judgment result through cross-modal reasoning to confirm whether the core entity in the visual scene strictly corresponds to the textual attribute in semantics.

Although deep learning has made significant progress in visual understanding and Natural Language Processing (NLP), existing methods still face some limitations when applied to street view images and POI consistency verification. First, many methods adopt a two-stage approach that separates Optical Character Recognition (OCR) from subsequent text matching. In such separated procedures, errors generated in the preceding recognition stages tend to propagate directly into the matching phase, thereby weakening the overall robustness of the

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system (Gomez and Karatzas, 2016; Wang et al., 2021). Secondly, street view images exhibit significant scale heterogeneity, as text regions associated with POI (e.g., shop signs) typically occupy only a small portion of the image. Under such conditions, generic visual models struggle to stably capture fine-grained textual features while suppressing complex background noise (Sun et al., 2019a; Chen et al., 2021). Third, the problem of cross-modal semantic misalignment remains insufficiently addressed. Most existing multimodal models emphasize global scene semantic alignment, while lacking fine-grained, entity-oriented alignment mechanisms, making it difficult to establish reliable semantic correspondences between visual patches and textual tokens (Maddalena et al., 2020; Yao et al., 2021).

To address these limitations, this paper proposes SignSpotter, a multi-scale vision-language framework designed for semantic consistency verification of geographic entities. The main contributions of this work are summarized as follows:

(1) This paper formalizes the matching process between street view images and POI data as an entity level semantic consistency discrimination problem. This problem framework requires the model to explicitly distinguish semantic consistency and conflict relationships under high similarity interference samples, which provides a problem modeling approach that more closely aligns with practical requirements for automated verification in geographic information quality control.
(2) A scene text informed visual pre training strategy is proposed to enhance the sensitivity of visual representations to text related visual clues in natural street scenes by pre training the visual backbone network on large-scale scene text datasets. This strategy effectively bridges the distribution gap between generic visual features and urban scene text visual features, thereby demonstrating that generic visual models can be effectively adapted to specific complex application scenarios through task driven pre training.
(3) A multi-scale asymmetric dual tower architecture and a cross-modal semantic reasoning mechanism are designed. The former jointly models the multi-scale visual features of street view images and the high-level semantic attributes of POIs. The latter establishes fine-grained alignment between visual regions and POI attributes through bidirectional interaction. This mechanism achieves a transition from shallow similarity matching to deep semantic reasoning, providing a reliable modeling framework for cross-modal semantic consistency discrimination of geographic entities.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the overall framework of SignSpotter and its key components. Section 4 reports the experimental setup and comparative evaluation results. Section 5 concludes the paper and discusses directions for future research.

2. Related work

2.1. POI data quality assessment and verification

POIs are fundamental units for characterizing urban spatial functions and constitute a core data foundation for LBS, urban computing, and navigation applications. The quality of POI data, particularly in terms of completeness, positional accuracy, and semantic consistency, directly determines the reliability and usability of geographic information services (Yeow et al., 2021; Yao, 2024). As a result, POI data quality assessment has long been a central research topic in GIS.

Early studies on POI verification primarily relied on geometric rules and statistical constraints. Typical approaches included detecting positional anomalies through spatial relationship analysis or identifying potential attribute conflicts by integrating trajectory data, social media, and other heterogeneous data sources (Hochmair et al., 2017; Touya et al., 2017; Klinkhardt et al., 2023). While these methods are effective in identifying logical inconsistencies or location deviations, their ability to verify semantic attributes of POIs remains limited. With the widespread adoption of VGI, street view imagery has gradually emerged as an important data source for POI semantic quality verification, as it provides a direct visual representation of the real physical environment (Biljecki and Ito, 2021). In recent years, a growing body of research has explored the use of street view images for tasks such as building function recognition, merchant existence detection, and facade feature extraction, in support of POI updating and maintenance (Kang et al., 2018; Sharifi Noorian et al., 2020; Liu et al., 2020). However, these approaches predominantly focus on macroscopic scene understanding and lack systematic modeling of fine-grained semantic consistency, particularly with respect to POI names and categories. More importantly, strong semantic cues in street view imagery that directly indicate POI identity, such as shop sign text, have not been fully exploited. This limitation significantly constrains the ability of existing methods to handle complex semantic errors, including name changes and category confusion.

2.2. Street view understanding and scene text recognition

Street view images represent a class of visual data that is highly unstructured, exhibits substantial scale variation, and contains complex background clutter. Early approaches to street view understanding mainly relied on handcrafted features for object detection and scene analysis, which showed limited effectiveness in complex urban environments (Girshick et al., 2014). With the development of deep learning, models based on Convolutional Neural Networks (CNN) and Transformers have achieved notable progress in the analysis of buildings, roads, and urban facilities (Chen et al., 2018; Xie et al., 2021).

In the context of POI semantic verification, the most discriminative visual cues in street view imagery often originate from scene text, such as shop signs and address plates. As a result, Scene Text Recognition (STR) technologies have been widely regarded as an important bridge connecting visual content with POI (Zhang et al., 2020). Existing methods for STR have achieved substantial advances in handling multi-oriented, curved, and complex background text (Shi et al., 2016; Zhou et al., 2017; Long et al., 2018). However, directly applying generic STR models to POI and street view image verification tasks still presents notable limitations. On the one hand, textual regions in street view images typically occupy only a small proportion of pixels, making single-scale feature extraction strategies prone to losing critical information (Ibrayim et al., 2022). On the other hand, most existing approaches adopt a two-stage approach in which text extraction and semantic matching are treated as independent steps. Critically, in such separated procedures, errors generated in the preceding recognition stages tend to propagate directly into the matching phase, thereby weakening the overall robustness of the system and its ability to perform end-to-end correction (Wang et al., 2021).

2.3. Cross-modal representation learning and matching

Cross-modal representation learning aims to construct a unified feature space where heterogeneous modalities such as vision and text can achieve effective alignment and similarity measurement. Early methods typically adopted a dual tower structure and were optimized through triplet loss or contrastive learning (Faghri et al., 2017). Subsequently, Vision-Language Pre-trained (VLP) models represented by CLIP and ALIGN established global semantic alignment capabilities through large-scale image text pair training and exhibited strong generalization in zero shot transfer (Jia et al., 2021; Radford et al., 2021). These models are primarily optimized for generic macroscopic scene semantics. When directly applied to tasks such as street view and POI verification, which highly rely on local details and dense scene text perception, their performance is often limited.

Building upon this, cross-modal representation learning research has gradually shifted from macroscopic alignment to fine-grained cross-modal perception, thereby deriving two mainstream research branches. On the one hand, scene text retrieval methods represented by OCR-free approaches attempt to directly establish the correspondence between vision and linguistic semantic text, thereby achieving end to end cross-modal modeling. For instance, from SPTS v2 (Liu et al., 2023), which early on verified the feasibility of end-to-end text recognition, to the recent FDP (Zeng et al., 2024) and CAYN (Qin et al., 2025), these works successfully adapted generic VLP models to scene text retrieval tasks without explicit transcription. Similarly, FastTCM (Yu et al., 2024) transformed CLIP into a scene text recognizer through scene text aware cross-modal adaptation. On the other hand, to address the issue that global representations struggle to characterize local details, fine-grained patch-token alignment mechanisms have been widely introduced. Inspired by the multimodal cross attention design in BLIP-2 (Li et al., 2023), current methods generally focus on constructing dense local feature correspondences. For example, models such as BOOM (Li et al., 2024), MULMAPPER (Jang and Chiang, 2024), and SpatialVLM (Chen et al., 2024) established an entity level semantic foundation in complex multimodal and geospatial tasks by calculating dense similarity matrices or employing local cross attention.

Although these advanced methods have achieved significant progress in their respective fields, they still face several adaptation challenges when directly applied to the semantic consistency verification of street views and POIs. Structurally, OCR-free retrieval models rely on single scale visual tokenization. When processing minute signboard text with drastic scale variations in street views, the stability of feature extraction might be limited. Regarding alignment mechanisms, current fine-grained frameworks mostly focus on soft similarity matching. When handling highly similar yet distinct POI entities, they lack an explicit reasoning mechanism targeting semantic conflicts. To this end, the proposed SignSpotter explicitly models this task as a binary classification matching problem. It not only introduces a multi-scale visual tower to alleviate scale heterogeneity but also employs a cross-modal reasoning module inspired by Natural Language Inference (NLI) to explicitly model semantic co-occurrence and conflicts by calculating feature differences and products, thereby achieving more precise consistency inference.

3. Method

3.1. Problem formulation and framework overview

To verify semantic consistency between street view images and POI data, this study formulates the consistency judgment between street view images and POI as a binary classification problem. Specifically, given a dataset $D\{(I_i, T_i, y_i)\}_{i=1}^{N}$ containing N samples, I_i denotes an original street view image that potentially contains merchant signage, and T_i represents a textual sequence composed of the POI name and category. The binary label $y_i \in \{0, 1\}$ indicates whether the image and the POI attributes are semantically consistent, where 1 denotes semantic consistency and 0 denotes inconsistency. It is particularly worth noting that, given the complexity of real-world map maintenance scenarios, the definition of semantic consistency in this study is not strict exact string matching but rather an entity-level verification based on the target POI as a reference. Field-captured merchant signboards may adopt prominent expressions such as abbreviations, short names, or omitted modifiers, whereas official POI records are typically more formal and comprehensive. Therefore, when $y_i = 1$, it indicates that the uploaded street view image genuinely presents the physical entity corresponding to the POI. Under this setting, the model tolerates reasonable expression differences, such as abbreviations or short names, between the image text and the official records, provided that the core semantics of the visual content point to the target entity. Conversely, when $y_i = 0$, it signifies that the street view image does not contain the target POI. Such negative samples encompass not only obvious errors where the image displays completely unrelated entities but also challenging cases where the image exhibits other easily confusable entities with highly similar names within the same category.

In practical map maintenance workflows, this reference-based verification task is typically deployed subsequent to a spatial coarse screening. The system first aggregates street view images into the candidate album of a specific POI based on spatial distance constraints and subsequently utilizes the proposed model for independent semantic verification. When confronting complex street view images containing multiple adjacent merchant signboards, the system simply combines the image with various surrounding candidate POIs to form independent data pairs. Because each model inference is guided by the specific target POI semantics as a query, it inherently filters out the visual interference of other irrelevant shops. This mechanism effectively resolves the matching difficulties associated with multiple shops in a single image and the redundant computations across multiple viewing angles.

The objective of this study is to learn a cross-modal mapping function $F_\theta(I, T)$, where θ denotes the learnable parameters of the model. This function is designed to predict the probability that a street view image and its associated POI attributes are semantically consistent, which is formulated as:

$$\widehat{y} = F_\theta(I, T) = P(y = 1 | I, T) \tag{1}$$

By minimizing the cross-entropy loss between the predicted probability $\widehat{y}$ and the ground truth label y, the model is trained to perform semantic consistency verification between street view images and POI data.

The overall architecture of SignSpotter consists of three cooperative modules, as illustrated in Fig. 1.

First, a scene text-informed visual pre-training module is introduced to enhance the ability of the visual backbone network to perceive fine-grained textual regions in street scenes. Second, a multi-scale asymmetric dual-tower representation module is constructed to separately extract multi-scale visual features from street view images and high-level textual features from POI attributes. Finally, a cross-modal semantic reasoning and consistency verification module is designed to enable semantic consistency inference between street view images and POI data.

3.2. Scene text-informed visual pre-training

Textual regions in street view images, such as shop signs and address plates, serve as key semantic anchors that connect visual scenes with POI data. However, general visual encoders typically focus on modeling global scene structure and have limited capabilities in perceiving fine-grained text details in local regions (He et al., 2016; Dosovitskiy, 2020; Liu et al., 2021). To address this limitation, this study proposes a scene text-informed visual pre-training strategy. Specifically, the pre-training stage is formulated as a STR task, in which the visual encoder is required to transcribe input signboard images into corresponding character sequences. Through strong character-level supervision, this process explicitly guides the model to focus on local structural patterns and discriminative texture features within textual regions. The overall workflow of the proposed pre-training strategy is illustrated in Fig. 2.

3.2.1. Pre-training data construction

To adequately cover the diversity of font styles, layout patterns, and imaging conditions encountered in real world street scenes, a large-scale hybrid scene text dataset is constructed by combining the ICDAR 2019 LSVT (Sun et al., 2019b) and ICDAR 2019 ReCTS (Zhang et al., 2019) datasets. Specifically, the LSVT dataset is collected from unconstrained real-world urban street scenes, exhibiting high scene complexity and multi-scale text distribution characteristics. The ReCTS dataset is

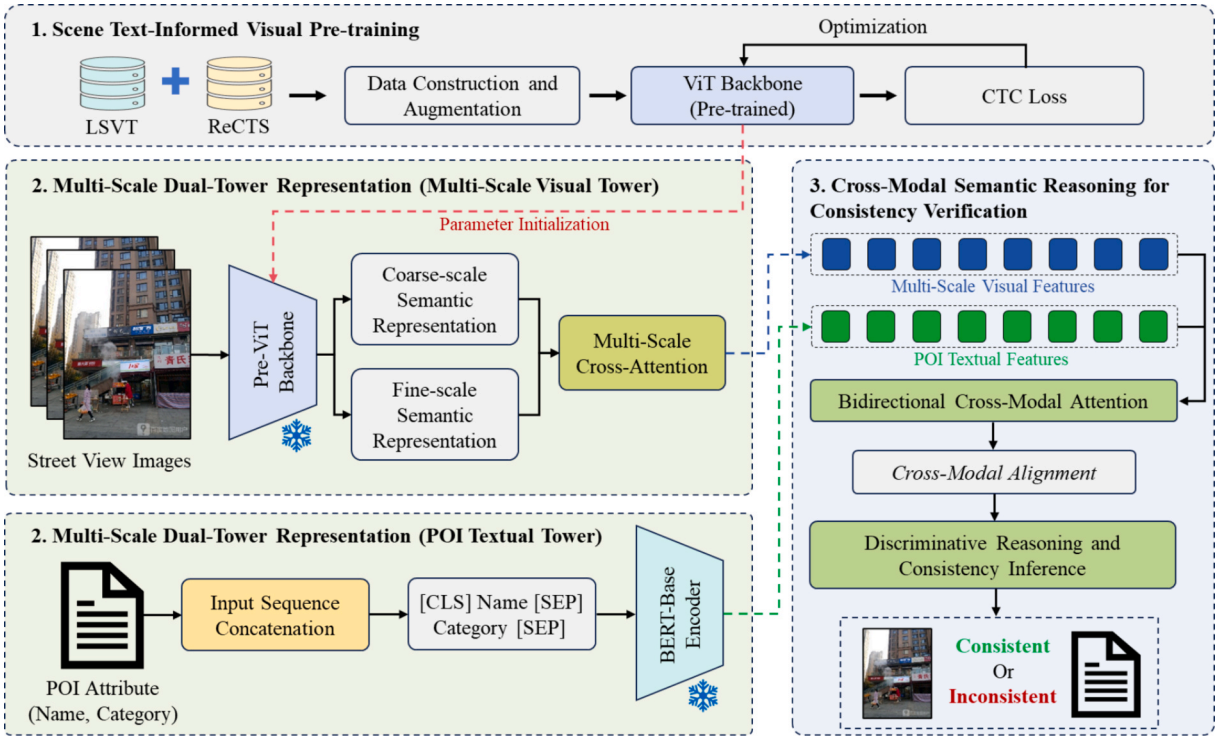


Fig. 1. Overall framework of SignSpotter.

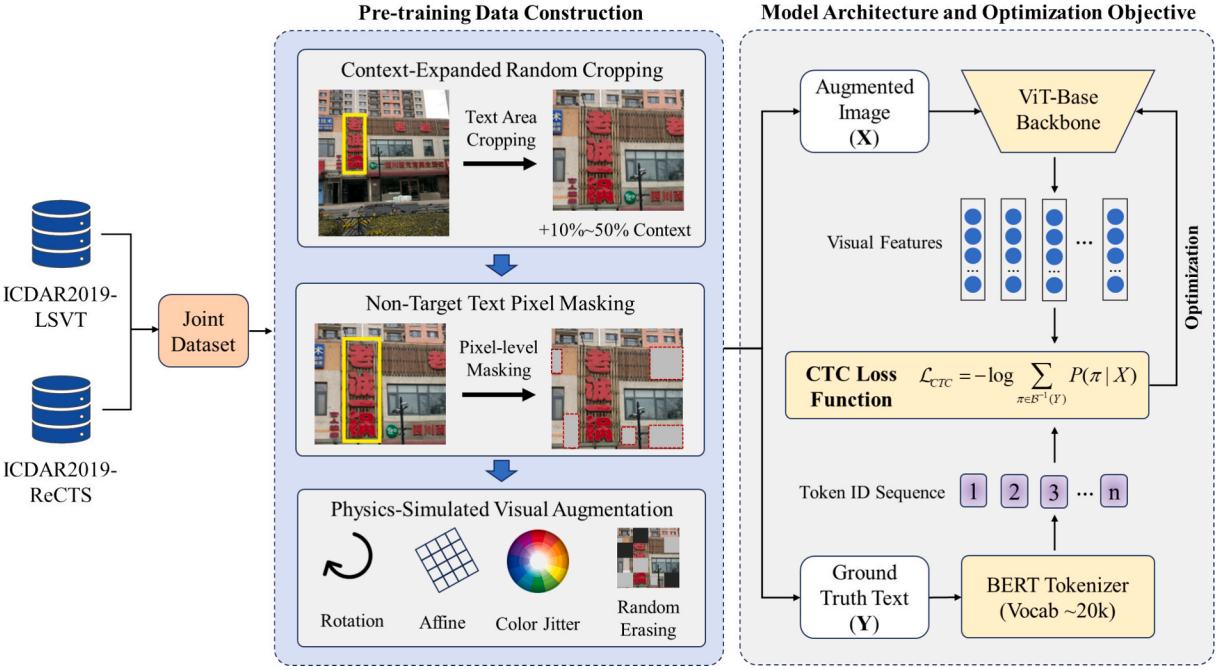


Fig. 2. Illustration of the scene text-informed visual pre-training.

specifically tailored to various complex merchant signboard designs, encompassing a rich variety of Chinese fonts alongside vertical and irregular layouts. Combining these two datasets not only endows the model with generalization capabilities in macroscopic complex street scenes but also injects domain-specific textual priors for the target POIs. Considering the complexity of street scene text in terms of scale variation, background interference, and imaging degradation, several data construction and augmentation strategies are further designed.

(1) **Context-Expanded Random Cropping.** Conventional cropping methods are prone to truncating character boundaries due to annotation inaccuracies or overly tight cropping regions. To mitigate this issue, a dynamic context expansion factor is introduced prior to cropping based on annotated bounding boxes. The original bounding box is randomly expanded by 10 to 50% in all directions. This strategy preserves local visual context surrounding the target text while effectively reducing the risk of

semantic truncation, thereby ensuring the integrity of character structural information.

(2) **Non-Target Text Pixel Masking.** To suppress background semantic interference caused by densely distributed signboards in street scenes, full annotation information is utilized to identify pixels corresponding to non-target text within the cropped region. These pixels are then filled with a global mean grayscale value. This operation suppresses interference from neighboring text while guiding the attention mechanism of the model to focus on the target character sequence, thus reducing feature confusion.

(3) **Physics-Simulated Visual Augmentation.** This strategy integrates both geometric and photometric perturbations. Geometric transformations include random rotation and affine shearing to simulate camera viewpoint deviations, while photometric perturbations include color jittering and random erasing to model illumination variation and partial occlusion. By introducing these perturbations, the model is encouraged to learn text representations that are robust to variations in imaging conditions, thereby improving generalization performance in complex open street scene environments.

3.2.2. Model architecture and optimization objective

In terms of model architecture, ViT-Base is adopted as the visual backbone network. Enhanced text images are divided into fixed size patch sequences and encoded into visual features (Dosovitskiy, 2020). To accommodate the supervision requirements of the STR task and to cover a large number of Chinese characters and special symbols commonly appearing on street signs, a vocabulary containing more than 20,000 characters is constructed using the BERT tokenizer (Devlin, 2018). Annotated text is then converted into corresponding token sequences.

To address the inconsistency in sequence length and alignment between input visual features and output character sequences in STR task, Connectionist Temporal Classification (CTC) is employed as the optimization objective (Graves et al., 2006). CTC optimizes the model by maximizing the conditional probability of the target character sequence Y given the input image X. The detailed mathematical formulation of the CTC loss is provided in Appendix A.

After pre-training, the learned weights of the visual encoder are transferred to the visual tower of SignSpotter and serve as initialization parameters for the subsequent multi-scale feature extraction module.

3.3. Multi-scale dual-tower representation

After obtaining the Pre-trained ViT (Pre-ViT), this study further constructs an asymmetric dual-tower representation architecture to separately model the multi-scale visual features of street view images and the textual features of POI.

3.3.1. Multi-scale visual tower

To simultaneously capture global scene context and local textual details, the visual tower is built upon a Pre-ViT backbone. As illustrated in Fig. 3, we design a resolution-adaptive mechanism that dynamically reconfigures feature extraction pathways based on the input resolution. Let I denote the input street view image, and $f_{enc}(\cdot)$ represent the Pre-ViT encoder. We construct a dual-pathway architecture comprising a fine-grained branch to extract local details and a coarse-grained branch to capture holistic structural semantics.

High-Resolution Strategy ($H, W \geqslant 224$). For high-quality images, we adopt a parallel modeling strategy. In the fine-grained branch, the image I is spatially partitioned into a 2×2 grid of local crops. Each crop is independently resized and encoded to form the local feature sequence v_{fine}, ensuring high-frequency details are preserved. Simultaneously, in the coarse-grained branch, the entire image I is resized and encoded to obtain the global context vector v_{coarse}. This design explicitly decouples local texture perception from global scene understanding.

Low-Resolution Strategy ($H, W < 224$). For low-quality inputs, we employ a Pseudo-Coarse Generation strategy to maintain structural consistency. In the fine-grained branch, to avoid fragmentation, we abandon partitioning and instead use direct up-sampling to preserve the original pixel information, and the feature sequence is padded with zero

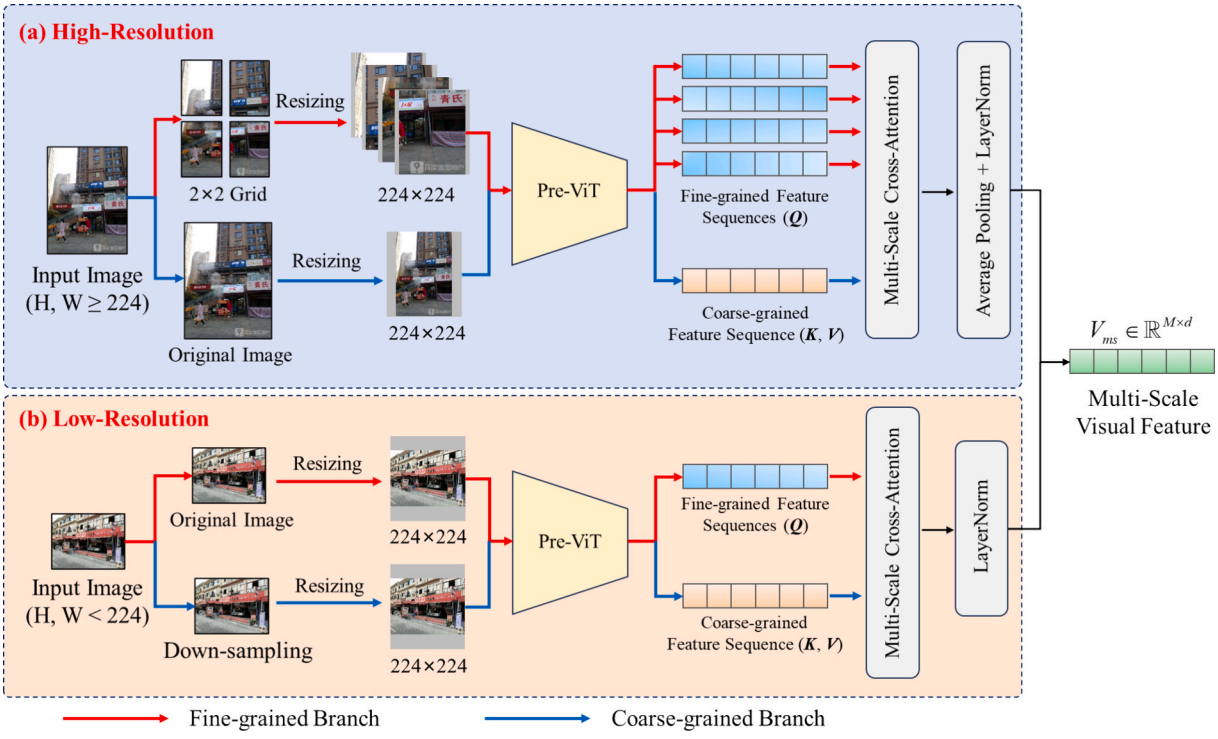


Fig. 3. Architecture of the multi-scale visual tower with resolution-adaptive encoding.

tensors to align dimensions. In the coarse-grained branch, to reconstruct stable structural priors from blurry inputs, we introduce a bottleneck transformation. The image I is first down-sampled by a factor of 2 to suppress high-frequency noise and then up-sampled before encoding. This process acts as a structural filter.

Detailed mathematical definitions for the aforementioned resizing, padding, and bottleneck operations can be found in Appendix A.

To fuse the complementary multi-scale features described above, a multi-scale cross-attention module is introduced at the end of the visual tower. In this module, the fine-grained feature sequence is treated as the Query (Q), while the coarse-grained feature sequence serves as the Key (K) and Value (V). The computation process is defined as follows:

$$Attention(Q,K,V) = Softmax\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{2}$$

$$V_{ms} = LayerNorm(Q + Attention(Q,K,V)) \tag{3}$$

where $d = 768$ denotes the hidden dimension.

For high resolution input, four local fine-grained feature sequences interact with the global feature sequence separately, and then are fused through average pooling. For low resolution inputs, direct interaction is performed between the fine-scale features and the pseudo coarse-grained features.

Finally, the visual tower outputs a unified multi-scale visual feature sequence $V_{ms} \in \mathbb{R}^{M\times d}$, where M denotes the number of visual patches. This representation provides a structurally consistent visual input for subsequent cross-modal semantic reasoning.

3.3.2. POI textual tower

Compared with visual information, POI textual attributes are relatively short in length but contain highly condensed entity-level semantic information. The objective of the textual tower is to map unstructured POI name and category attributes into dense representations with high-level semantic constraints. Specifically, the POI name and category are concatenated into an input sequence in the form of [CLS] Name [SEP] Category [SEP]. This sequence is converted into a token sequence T_{in} using the BERT tokenizer and then fed into a pre-trained BERT model (Devlin, 2018), which outputs a textual feature sequence:

$$T_{sem} = BERT(T_{in}) \in \mathbb{R}^{L\times d} \tag{4}$$

where L denotes the number of textual tokens and $d = 768$ denotes the hidden dimension.

This textual feature not only encodes the literal semantics of the POI name but also implicitly models the semantic constraints imposed by the category attribute through the [SEP] structure. For example, the strong semantic association between a specific brand name and its corresponding business category is captured within the contextual representation. As a result, the textual tower provides a semantically complete and highly discriminative textual feature for subsequent cross-modal alignment.

3.4. Cross-modal semantic reasoning for consistency verification

In the preceding modules, the model separately obtains a multi-scale visual feature sequence from street view images, denoted as V_{ms}, and a textual feature sequence from POI attributes, denoted as T_{sem}. However, due to the heterogeneity in expression form and spatial distribution between visual and textual data, a clear semantic correspondence has not yet been established between visual patches and text tokens. To bridge this semantic gap and achieve entity-level consistency verification, this paper designs a cross-modal semantic reasoning module, whose overall framework is shown in Fig. 4.

3.4.1. Cross-modal alignment in shared semantic space

This study first projects the visual feature sequence and the textual feature sequence into a unified latent semantic space through linear projection layers. On this basis, a bidirectional cross-modal multi-head attention mechanism is introduced to explicitly model fine-grained associations between visual regions and textual keywords. Specifically, the model uses textual features as queries to retrieve relevant visual fea-

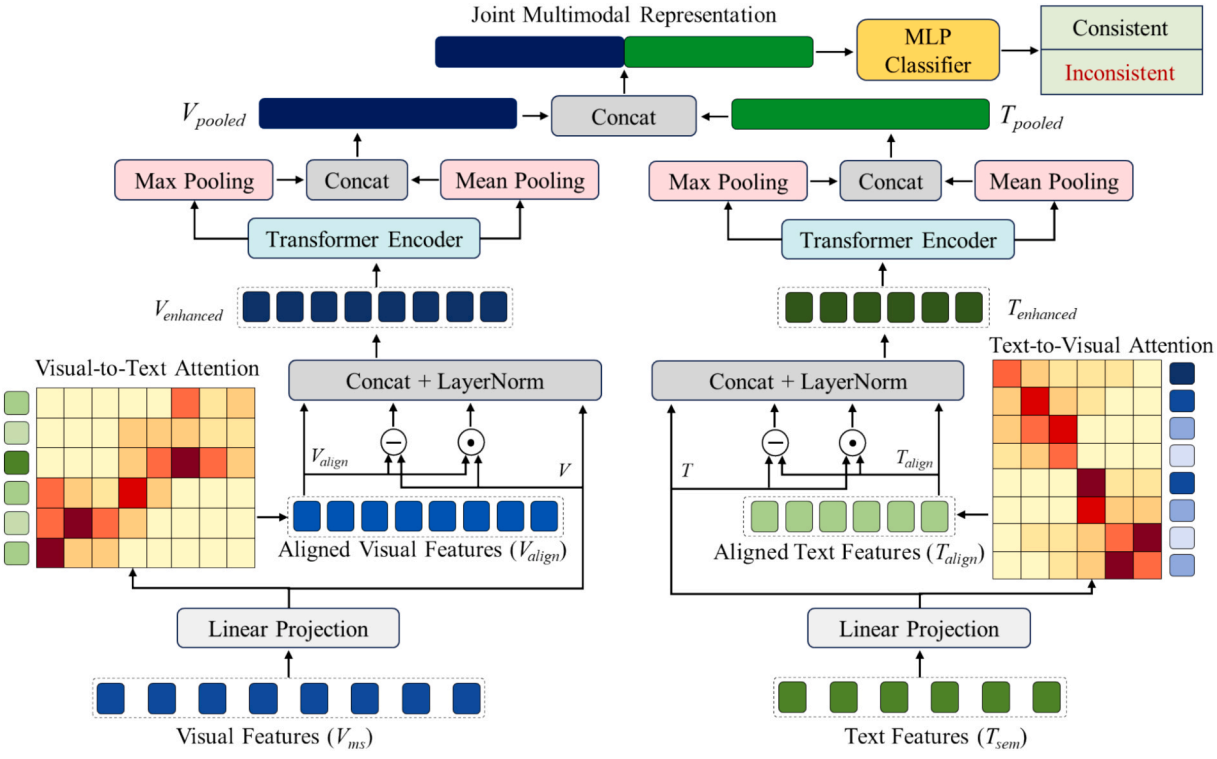


Fig. 4. Cross-modal semantic reasoning and consistency verification module of SignSpotter.

tures, while simultaneously using visual features as queries to retrieve relevant textual features. Through this bidirectional retrieval process, soft alignment between visual patches and textual tokens is achieved. The alignment process can be formulated as follows:

$$V_{align} = MultiHead(V, T, T) \tag{5}$$

$$T_{align} = MultiHead(T, V, V) \tag{6}$$

where V_{align} and T_{align} denote the aligned cross-modal feature representations. The symbols V and T represent the projected visual and textual features, respectively. $MultiHead(\cdot)$ refers to the standard multi-head attention mechanism (detailed in Appendix A).

3.4.2. Discriminative reasoning and consistency inference

Relying solely on attention alignment is insufficient to fully support the discriminative requirements of semantic consistency verification, particularly when identifying samples that exhibit semantic conflicts. Inspired by NLI tasks, we construct a discriminatively enhanced representation based on the cross-modal aligned features to explicitly model the semantic consistency and inconsistency relationships between modalities (Chen et al., 2016). Specifically, element-wise difference and element-wise product operations are employed to characterize semantic conflict and semantic co-occurrence, respectively.

$$V_{enhanced} = LayerNorm\big(V_{align} + W_v^f \bullet Concat\big(V, V_{align}, V - V_{align}, V \odot V_{align}\big)\big) \tag{7}$$

$$T_{enhanced} = LayerNorm\big(T_{align} + W_t^f \bullet Concat\big(T, T_{align}, T - T_{align}, T \odot T_{align}\big)\big) \tag{8}$$

where W_v^f and W_t^f denote learnable weight matrices, and $\odot$ denotes element-wise multiplication. The difference terms amplify semantic conflicts between modalities. For example, when an image displays the sign "KFC" while the associated text corresponds to "McDonald", the difference component produces strong responses. In contrast, the product terms reinforce evidence of semantic consistency.

Meanwhile, the enhanced visual and textual feature sequences are fed into multi-layer Transformer encoders for semantic composition. Subsequently, a hybrid pooling strategy (concatenating max and average pooling) is adopted to map variable-length sequences into fixed-length global representations. Finally, they are concatenated and fed into a fully connected network for semantic consistency prediction. The model is trained by minimizing the cross-entropy loss between the predicted results and the ground truth labels. The formal definitions of the pooling operations and the classifier are detailed in Appendix A.

4. Experiments

This section presents a comprehensive experimental evaluation of SignSpotter with respect to its effectiveness and robustness in the task of semantic consistency verification between street view images and POI data. We first introduce the datasets and evaluation metrics used in the experiments. The performance of the proposed method is then analyzed from three perspectives, including comparative experiments, cross city transfer experiments, and robustness analysis under visual occlusion. Finally, ablation studies and qualitative visualizations are conducted to validate the contribution of each key component.

4.1. Datasets and evaluation metrics

4.1.1. Dataset Description

To comprehensively evaluate the semantic consistency verification capability of the proposed model in complex real world street scene environments, a multi-level dataset system is constructed and used, covering pre-training dataset, source city dataset, and target city dataset.

(1) **Pre-training Dataset:** In the scene text-informed visual pre-training phase, as described in Section 3.2.1, the ICDAR 2019 LSVT and ICDAR 2019 ReCTS datasets are jointly utilized, compiling a total of approximately 50,000 training images. By conducting supervised STR pre-training on this corpus, the model effectively learns robust character textures and local structural features.

(2) **Source City Dataset:** The core experimental data of this study originates from the Baidu Maps platform in Chengdu, China. The fundamental attributes of the POIs, such as name, category, address, and coordinates, have undergone official verification on this platform. Furthermore, the user-uploaded street view images generally exhibit high authenticity and objective correspondence, providing reliable data support for this research. However, considering the diverse types of user-uploaded images, they inevitably contain noisy data such as indoor environments and close-up product shots. Therefore, to construct a high-quality dataset, we designed a positive and negative sample construction pipeline that integrates automated retrieval and manual verification.

During the construction of positive samples, for each target POI, we first randomly select a street view image from its candidate album and extract all the text within the image using OCR technology. Subsequently, the edit distance is utilized to calculate the similarity S between the extracted text and the current POI name. If the similarity $S \geq 0.8$, the image is automatically accepted as a high-confidence positive sample, and the retrieval process for the current POI is immediately terminated. If this threshold is not met, the next image in the album is randomly sampled to repeat the aforementioned calculation. If all images in the album have been traversed and no image exceeds the similarity threshold, the image with the highest similarity score, denoted as S_{max}, across the entire album is selected for the final judgment. If this score falls within the medium-confidence interval, namely $0.4 \leq S_{max} < 0.8$, it indicates a partial character overlap, and such samples are handed over to annotators with professional backgrounds for manual review. If the score belongs to the low-confidence interval, namely $S_{max} < 0.4$, it is determined that no scene text image matching the target POI exists in the album, thereby directly discarding this POI record. In practical data processing, a total of 241,567 POI records satisfied the high-confidence condition, 32,178 records fell into the medium-confidence interval, and 10,516 records were discarded due to exceedingly low similarity. Finally, to ensure a balanced number of positive and negative samples for subsequent experiments, we proportionally sampled 120,000 positive samples according to the ratio of each POI category in the original data pool, thereby preserving the authentic spatial distribution characteristics of urban POIs.

During the construction of negative samples, we employed a Hard Negative Sampling strategy to generate an equal number of negative samples corresponding to the acquired 120,000 positive samples, maintaining a positive and negative ratio of 1:1. Specifically, when generating the corresponding negative sample for each positive sample, we kept the fundamental attributes of the target POI, such as name and category, unchanged and constructed negative sample pairs by matching them with inconsistent street view images. When selecting replacement images, we prioritized extracting images from other POI records that share the same category and exhibit highly similar names to the target POI to construct hard negative samples. This discriminative process followed the previously established similarity metric, requiring that the textual similarity S between the source POI providing the image and the target POI satisfies $S \geq 0.6$. Such samples are utilized to evaluate the fine-grained discriminative capability of the model at the entity level, encompassing cases like chain stores sharing the same name but featuring different branch modifiers or distinct shops with similar

names. If the high similarity condition could not be met within the data pool, images were extracted from POIs within the same category but possessing different names, satisfying $0.3 \leq S < 0.6$, to construct medium negative samples. If the aforementioned matching conditions within the same category could not be satisfied, images were randomly sampled from the remaining POIs across different categories to construct easy negative samples. In this scenario, the source POI providing the image and the target POI exhibited no semantic overlap in both category and name. This strategy ensured the diversity of negative samples while effectively avoiding an overly simplistic negative sample distribution during the training process. In the finally constructed negative sample set, the proportions of easy, medium, and hard negative samples were approximately 25%, 33%, and 42%, respectively.

Through the aforementioned pipeline, we ultimately constructed a source city dataset comprising 120,000 pairs of positive and negative samples. The macroscopic statistics and distribution characteristics of this dataset are presented in Table 1. Regarding the category distribution, the dataset encompassed 19 primary industry categories. It should be noted that this merging operation was solely utilized for data visualization purposes. During the actual computation and training stages, the model still utilized the original authentic categories. In terms of textual features, over 90% of the POI name lengths were distributed within 17 characters, exhibiting significant attributes of short texts and high information density. Such highly condensed texts not only align with the signboard characteristics in real world urban scenarios but also imply that minor character variations can induce substantial semantic shifts. This further corroborates the necessity of employing the previously mentioned hard negative sampling strategy to enhance the fine-grained perceptual capability of the model. Finally, to conduct subsequent model training and performance evaluation, the constructed 120,000 sample pairs were partitioned into a training set of 100,000 pairs, a validation set of 10,000 pairs, and a testing set of 10,000 pairs.

(3) **Target City Dataset:** To evaluate the performance of the proposed model across different geospatial regions, a target city test set was constructed. The construction pipeline for the positive and negative samples in this dataset is identical to that of the source dataset. The test set comprises street view images and POI data sourced from Baidu Maps for Beijing, Shanghai, Guangzhou, and Xi'an. Each city contains 2,000 positive sample pairs and 2,000 negative sample pairs. These data are exclusively utilized for final testing and are completely excluded from the training process.

To verify the transferability of the model across different platforms and data sources, an independent test set based on the Dianping platform was additionally constructed. This dataset covers the four target cities, with each containing 2000 positive and 2000 negative sample pairs. Unlike the macroscopic street views of Baidu Maps, Dianping images are primarily shop photos spontaneously uploaded by consumers, featuring more prominent visual subjects. Incorporating this dataset enables a comprehensive evaluation of the generalization performance of the model under diverse entity densities and image qualities.

Table 1
Statistical characteristics of the source city dataset.

POI category	Proportion	Name length (Characters)	Proportion
Shopping & Retail	32.14%	2–5	18.80%
Companies & Enterprises	12.86%	6–8	15.35%
Life Services	10.21%	9–12	35.05%
Beauty Services	7.11%	13–16	21.71%
Education & Training	5.40%	17–20	6.35%
Auto Services	5.23%	>20	2.74%
Food	3.81%	Average	10.45
Others	23.24%	Max Length	58

Note: The level-1 industry categories are defined according to the official classification system of Baidu Maps. The "Others" category includes the remaining 12 level-1 industry classifications with smaller proportions: Tourist Attractions (3.52%), Leisure & Entertainment (3.42%), Transportation Facilities (3.19%), Real Estate (2.25%), Culture & Media (2.11%), Hotels (2.06%), Entrances & Exits (1.65%), Finance (1.51%), Sports & Fitness (1.25%), Natural Features (1.20%), Healthcare (1.02%), and Government Institutions (0.04%).

4.1.2. Evaluation metrics

To comprehensively evaluate model performance, Precision (P), Recall (R), and the F1-score (F1) are adopted as evaluation metrics. Their definitions are given as follows:

$$P = \frac{TP}{TP + FP} R = \frac{TP}{TP + FN} \quad \text{F1} \;=\; 2 \times \frac{P \times R}{P + R} \tag{9}$$

where TP (True Positive) denotes the number of samples correctly identified by the model as "Consistent"; FP (False Positive) denotes the number of "Inconsistent" samples incorrectly classified as "Consistent"; FN (False Negative) denotes the number of "Consistent" samples missed and misclassified as "Inconsistent".

4.2. Implementation details

The proposed SignSpotter framework is implemented in PyTorch and trained and evaluated on a workstation equipped with a single NVIDIA RTX 4090 GPU. The overall training procedure follows a two-stage strategy to jointly balance scene text representation learning and downstream verification performance. In the first stage, the model is trained on a combined dataset consisting of LSVT and ReCTS. In the second stage, the parameters of both the visual and textual backbone networks are frozen, and only the multi-scale visual fusion module and the cross-modal reasoning head are fine-tuned. The key hyperparameter settings for both stages are summarized in Table 2.

4.3. Performance evaluation

4.3.1. Comparative experiments

We conduct quantitative comparisons on the Chengdu dataset against a diverse set of representative baseline methods, including OCR based two-stage matching approaches, traditional dual-tower models, general VLP models, and state-of-the-art methods specifically designed for street view or scene text tasks. All methods are trained and evaluated under identical data splits and evaluation metrics. The experimental results are summarized in Table 3.

Table 3 shows that SignSpotter achieves the best performance across all evaluation metrics, with Precision, Recall, and F1-score reaching 90.60%, 91.32%, and 90.96%, respectively. Compared with existing methods, SignSpotter substantially reduces the rate of false matches while maintaining a high Recall, demonstrating stronger cross-modal discriminative capability and overall robustness.

Table 2
Key implementation details of SignSpotter.

Category	Parameter	Value
Architecture	Visual Backbone	ViT-B/16
	Textual Backbone	BERT-base-Chinese
	Hidden Dimension	768
Input	Image Resolution	224 × 224
	Patch Size	16 × 16
	Max Text Length	20
Stage 1: Pre-training	Batch Size	64
	Learning Rate	3×10^{-5}
	Epochs	100
Stage 2: Downstream	Batch Size	128
	Learning Rate	1×10^{-4}
	Epochs	30
	Dropout Rate	0.1

Table 3
Quantitative comparison results of different methods on the Chengdu dataset.

	Methods	Precision (%)	Recall (%)	F1-score (%)
1	OCR + Edit Distance (Levenshtein, 1966; Du et al., 2020)	94.04%	70.08%	80.31%
2	OCR + BERT (Devlin, 2018; Du et al., 2020)	68.77%	92.38%	78.85%
3	ResNet-50 + BERT (He et al., 2016; Devlin, 2018)	70.19	81.97	75.62
4	CLIP (Radford et al., 2021)	73.40	83.28	78.03
5	ViLT (Kim et al., 2021)	60.33	88.29	71.68
6	BLIP-2 (Li et al., 2023)	74.30	85.94	79.70
7	MULMAPPER (Jang and Chiang, 2024)	89.71	69.72	78.46
8	BOOM (Li et al., 2024)	81.47	83.63	82.54
9	FastTCM (Yu et al., 2024)	81.54	88.77	85.00
10	CAYN (Qin et al., 2025)	76.02	88.51	81.79
11	**SignSpotter**	**90.60**	**91.32**	**90.96**

From the perspective of OCR based two-stage matching methods, different textual similarity calculation strategies lead to obvious performance differences. The Precision of the baseline method incorporating edit distance reaches as high as 94.04%, but its Recall is only 70.08%. This indicates that literal based matching rules lack sufficient flexibility and fault tolerance. Once OCR recognition errors or shop aliases occur, this method is highly prone to missed detections. To alleviate this issue, the subsequent baseline introduces the pre-trained BERT model for deep semantic matching. This strategy increases the Recall to 92.38%, but its Precision drops significantly to 68.77%. This demonstrates that relying solely on textual semantics easily introduces a larger number of false positive samples. The model struggles to effectively distinguish interfering items that are literally similar but represent different entities.

Analyzing the traditional dual-tower methods, the ResNet-50 + BERT baseline relies primarily on global visual features and text representations for matching. Lacking the capability to model local text regions and cross-modal interactions, it yields an F1-score of only 75.62%. General VLP models (e.g., CLIP and ViLT) exhibit relatively high Recall, but suffer from significantly limited Precision, with ViLT achieving only 60.33%. This indicates that while they possess a degree of semantic coverage, they are prone to misclassification and fail to effectively distinguish between POIs that are semantically similar but distinct entities. BLIP 2 outperforms the above methods in overall performance, reflecting the benefits of large-scale vision-language pre-training for cross-modal understanding. However, since its design objective targets general vision-language tasks, it exhibits limited adaptability when applied to POI scenarios characterized by highly localized visual cues, strong background noise, and heavy reliance on textual evidence in street view imagery.

Specialized methods designed for street view or scene text tasks, including MULMAPPER, BOOM, FastTCM, and CAYN, demonstrate strengths on individual metrics. For example, MULMAPPER achieves relatively high Precision, but its Recall is clearly limited, indicating insufficient coverage in cases involving missing text or diverse semantic expressions. BOOM improves the stability of cross-modal alignment through bidirectional consistency constraints, yet it remains less effective at suppressing semantically similar but irrelevant POIs under complex street scene backgrounds. FastTCM and CAYN perform well in terms of Recall, but their ability to suppress false matches is limited, resulting in overall F1-scores that remain below that of SignSpotter.

In contrast, SignSpotter achieves a more balanced performance between Precision and Recall. This advantage can be primarily attributed to the combined effects of scene text-informed visual pre-training, multi-scale visual feature modeling, and cross-modal semantic reasoning. Together, these components enable SignSpotter to simultaneously attend to key textual cues, local visual details, and global environmental context in street view imagery, leading to more robust semantic consistency verification under complex real-world scenarios.

In summary, SignSpotter possesses stronger cross-modal semantic modeling capabilities and discriminative robustness in complex street view environments, significantly outperforming existing mainstream methods in the task of verifying consistency between street view images and POI data.

4.3.2. Transfer experiments

Following the data construction protocol described in Section 4.1.1, the model is trained exclusively on the source city dataset from Chengdu and then directly transferred to four target city datasets, namely Beijing, Shanghai, Guangzhou, and Xi'an for testing. The aforementioned test data encompass two independent platforms, Baidu Maps and Dianping, and the quantitative results are presented in Table 4.

In the cross city transfer results on the Baidu Maps platform, although the model was trained solely on the Chengdu dataset, SignSpotter maintained a stable and high-level of performance across all four target cities. The F1-score for Beijing, Shanghai, Guangzhou, and Xi'an reached 86.44%, 86.54%, 89.00%, and 88.89%, respectively. Compared to the source city (90.96%), the performance decline was limited. These results demonstrate that the model possesses strong cross-city generalization capabilities when confronting diverse urban environments. Further analysis of the Precision and Recall metrics reveals that SignSpotter did not exhibit significant performance imbalance across the target cities. This suggests that the model does not overly rely on specific visual background patterns or text distribution characteristics unique to the source city. Notably, on the Guangzhou and Xi'an datasets, the model achieved performance levels approaching those of the source city. This indicates effective adaptability to variations in architectural styles, streetscape appearances, and signboard designs across different cities.

In the cross platform transfer results on the Dianping platform, SignSpotter similarly exhibits robust performance. In the majority of the tested cities, the various metrics for Dianping are even slightly higher than the corresponding Baidu Maps data for the same city. For instance, the Dianping F1-scores for Beijing and Guangzhou reach 87.32% and 89.92% respectively. This objective data phenomenon directly corroborates the differences in visual entity density between the two platforms. Baidu street views typically contain dense complex entities alongside background interference, whereas Dianping images predominantly focus on a single shop with much clearer scene text. The quantitative results demonstrate that SignSpotter can precisely capture such high-quality local scene text features and accomplish the matching process. This fully proves that the model can not only process macroscopic and dense street view images but also adapt to real world consumer scene photos with a microscopic focus.

In summary, the cross city and cross platform transfer experiments jointly verify the high robustness of SignSpotter. The model can maintain an extremely stable capability for semantic consistency discrimination on data from unseen new cities and new platforms. This highlights its immense practical application potential for cross regional deployment and automated verification tasks of large-scale entity data within complex real-world environments.

4.3.3. Robustness analysis against visual occlusion

To evaluate the robustness of SignSpotter when confronting the loss of key visual information (e.g., occlusion or blur) in real-world street view environments, we designed a targeted text region occlusion experiment. Unlike random occlusion applied to the entire image, this experiment first precisely localizes the text regions corresponding to the POI name within the image. Subsequently, random pixel masks of varying proportions are applied exclusively within these text regions. This approach more authentically simulates scenarios where textual information is missing due to physical occlusion, varying shooting angles, or environmental interference in street scenes. Specifically, we established four increasing occlusion intensities (Ratio = 0.1, 0.3, 0.5,

Table 4
Quantitative results of transfer experiments on four target city datasets across different POI platforms.

	City	Precision (%)		Recall (%)		F1 (%)	
		Baidu	Dianping	Baidu	Dianping	Baidu	Dianping
1	Beijing	84.94	85.88	88.00	88.80	86.44	87.32
2	Shanghai	84.34	85.20	88.85	89.20	86.54	87.15
3	Guangzhou	88.17	89.24	89.85	90.40	89.00	89.92
4	Xi'an	88.15	88.29	89.65	89.00	88.89	88.65
5	**Chengdu**	**90.60**	**91.14**	**91.32**	**91.70**	**90.96**	**91.42**

0.7) and used the non-occluded scenario (Ratio = 0.0) as the control group. Visual examples under different occlusion ratios are shown in Fig. 5. All experiments were conducted using the same model parameter configuration, and the quantitative results are summarized in Table 5.

As observed in Table 5, the overall performance of the model exhibits a smooth and reasonable decline as the occlusion ratio of POI-related text regions increases. Under mild occlusion conditions (Ratio = 0.1), SignSpotter maintains high discriminative performance with an F1-score of 86.57%, representing only a limited decrease compared to the non-occluded scenario (90.96%). This indicates that the model possesses strong robustness when facing partial text loss. When the occlusion ratio increases to 0.3 and 0.5, performance declines further, yet Precision and Recall do not show severe imbalance. This suggests that even with partial loss of key textual information, SignSpotter can still leverage remaining visible text, the overall storefront structure, and street view contextual information to perform cross-modal semantic reasoning, thereby preventing rapid performance degradation.

Table 5
Quantitative results of models under different visual occlusion ratios.

	Ratio	Precision (%)	Recall (%)	F1-score (%)
1	0.1	87.74	85.42	86.57
2	0.3	80.96	72.84	76.69
3	0.5	78.40	62.20	69.37
4	0.7	74.18	54.12	62.58
5	**0.0**	**90.60**	**91.32**	**90.96%**

Under high occlusion conditions (Ratio = 0.7), where most text regions directly related to the POI are randomly masked, the discriminative difficulty increases significantly, causing the F1-score to drop to 62.58%. Nevertheless, the model does not suffer a catastrophic collapse; Precision remains at 74.18%, while Recall drops to 54.12%. This

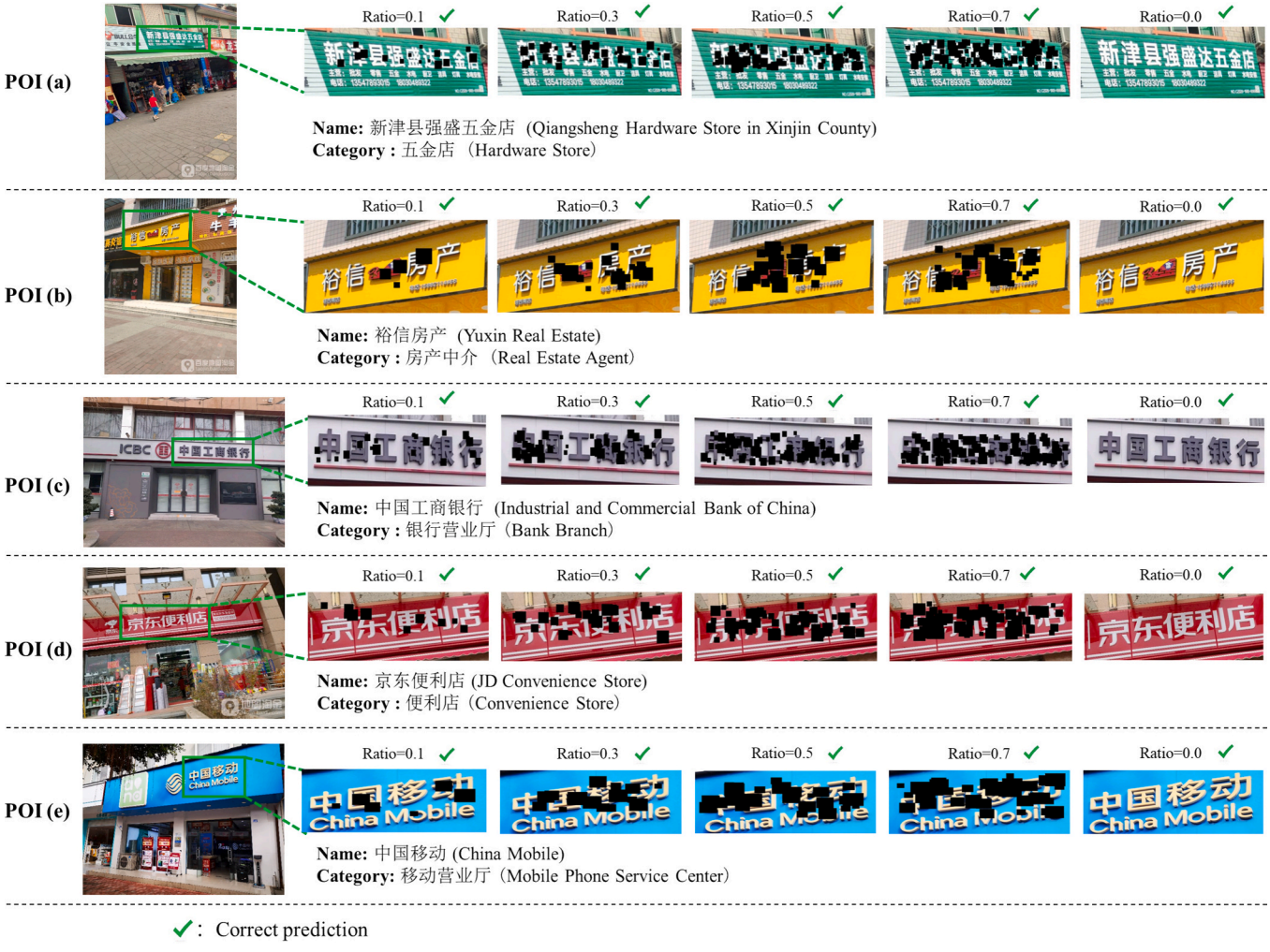


Fig. 5. Visual example of applying random occlusions of different proportions to text regions.

demonstrates that SignSpotter does not rely solely on explicit text recognition results for verification. Instead, in cases of severe textual information deficit, it can still comprehensively utilize non-textual visual cues, such as signboard morphology, color layout, architectural structure, and surrounding environmental semantics, to conduct supplementary reasoning.

The visualization examples in Fig. 5 further illustrate how the proposed framework verifies semantic consistency under severe text occlusion. In the case of POI (a), although the primary storefront name is largely occluded, secondary textual information describing business scope (e.g., product categories and services) remains partially visible. These category-related textual cues, together with consistent storefront layout and visual context, enable the model to correctly infer semantic consistency even in the absence of explicit POI name information. For POI (b), the model maintains correct predictions by combining partially visible characters with surrounding visual patterns, indicating that the model can utilize incomplete but discriminative textual cues. In contrast, POIs (c), (d), and (e), which correspond to standardized chain brands, exhibit robust predictions across all occlusion ratios. This robustness can be attributed to stable brand-level visual patterns, including logo structure, color schemes, and facade layouts, which provide reliable discriminative evidence for semantic consistency verification when textual regions are heavily occluded.

Overall, these cases demonstrate that the proposed framework does not rely solely on explicit text recognition, but instead flexibly combines residual textual cues, structural visual features, and contextual visual patterns to support robust entity-level semantic consistency verification in complex street view environments.

4.3.4. Case study

To further intuitively analyze the discriminative differences between varying methods in complex street view scenarios, we selected several representative samples for case analysis. As shown in Fig. 6, the selected examples cover four types of challenging street view situations common in practical applications, including semantic interference, degraded imaging conditions, small-scale targets, and occlusion. The figure also

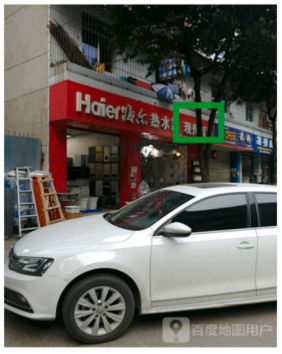
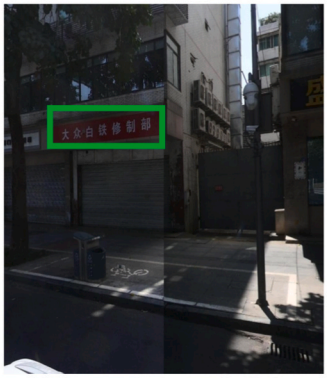
	(a) Semantic Interference	(b) Degraded Imaging Conditions	(c) Small-scale Targets	(d) Occlusion
Positive samples	 **Name:** 东方家具 (Oriental Furniture) **Category:** 家具店 (Furniture Store)	 **Name:** 小刘蔬菜店 (Xiao Liu's Vegetable Shop) **Category :** 菜店 (Vegetable Shop)	 **Name:** 优幼童趣馆 (Excellent Children's Fun Center); **Category :** 早教 (Early Childhood Education)	 **Name:** 现代灯饰 (Modern Lighting Fixtures) **Category :** 灯具 (Lighting Fixtures)
BOOM	Correct prediction	Incorrect prediction	Correct prediction	Incorrect prediction
FastTCM	Correct prediction	Incorrect prediction	Incorrect prediction	Correct prediction
CAYN	Incorrect prediction	Correct prediction	Incorrect prediction	Correct prediction
SignSpotter	Correct prediction	Correct prediction	Correct prediction	Correct prediction
	(a) Semantic Interference	**(b) Degraded Imaging Conditions**	**(c) Small-scale Targets**	**(d) Occlusion**
Positive samples	 **Name:** 卢氏布庄 (Lu's Fabric Store) **Category :** 家纺店 (Home Textile Store)	 **Name:** 大众白铁修治部 (Public Sheet Metal Repair Department) **Category :** 生活服务 (Life Service)	 **Name:** 南区运动场 (South District Sports Ground) **Category :** 体育场馆 (Sports Stadium)	 **Name:** 忠君好房 (Zhongjun Real Estate) **Category :** 房产中介 (Real Estate Agent)
BOOM	Incorrect prediction	Incorrect prediction	Correct prediction	Incorrect prediction
FastTCM	Correct prediction	Incorrect prediction	Incorrect prediction	Incorrect prediction
CAYN	Incorrect prediction	Correct prediction	Correct prediction	Incorrect prediction
SignSpotter	Correct prediction	Correct prediction	Correct prediction	Correct prediction

Fig. 6. Qualitative comparison under four challenging street view scenarios.

presents the prediction results of BOOM, FastTCM, CAYN, and the proposed SignSpotter to compare the discriminative behaviors of different models under identical visual inputs.

In the semantic interference scenario (Fig. 6(a)), street view images often simultaneously contain multiple adjacent shops or prominent textual elements, making the visual cues of the target POI susceptible to interference from irrelevant semantics. It can be observed that comparison methods are easily misled by neighboring text or highly salient visual elements in such scenarios, resulting in erroneous matches. In contrast, SignSpotter accurately completes the verification, indicating that its cross-modal attention mechanism achieves selective alignment at the visual patch and textual token levels, effectively suppressing irrelevant semantic interference unrelated to the target POI.

In the degraded imaging conditions scenario (Fig. 6(b)), factors such as motion blur, strong light reflection, or night-time noise make POI-related text difficult to recognize clearly. Under these circumstances, some baseline methods fail to establish stable cross-modal correspondences due to their strong reliance on explicit text regions or clear character structures, leading to verification failure. SignSpotter, however, still provides correct predictions. This demonstrates that it does not rely solely on character-level textual information but can comprehensively utilize the overall signboard morphology, color style, and contextual visual structure for semantic reasoning, thereby maintaining good robustness under degraded imaging conditions.

In the small-scale targets scenario (Fig. 6(c)), the target POI occupies a small proportion of the entire street view image, and the resolution of the storefront and text regions is limited, making them prone to being overwhelmed by complex backgrounds. Experimental results show that most comparison methods exhibit varying degrees of missed detection in such samples, whereas SignSpotter successfully completes the prediction. This success is attributed to its multi-scale visual modeling strategy, which enables the model to capture local visual cues highly relevant to the POI text within the macroscopic background, thereby mitigating the discriminative difficulties caused by the small scale of the targets.

In the Occlusion scenario (Fig. 6(d)), key visual areas of the POI are partially obstructed by pedestrians, vehicles, or other foreground objects, resulting in an incomplete presentation of storefront or textual information. It can be observed that the prediction results of some comparison methods are unstable under occlusion conditions, while SignSpotter maintains correct predictions. This indicates its ability to effectively integrate residual visible text, local structural features, and overall scene context information during the cross-modal alignment process, establishing stable semantic associations even under incomplete information conditions.

To further quantitatively verify the model's robustness under challenging scenarios, we randomly sampled 50 instances for each of the four typical complex scenarios from the test set to construct a class-balanced hard-example subset. As shown by the quantitative results in Fig. 7(a), under these extreme conditions, the performance of the baseline methods degraded to varying degrees. In contrast, our proposed SignSpotter maintained the optimal F1-score across all scenarios. This quantitative data not only strongly corroborates the qualitative visual observations in Fig. 6, but also further confirms the efficiency and stability of the proposed architecture in handling complex semantic interference and local information missing.

Despite SignSpotter demonstrating strong robustness in most complex street view scenarios, misjudgments still occur under certain extreme conditions. Fig. 8 presents several representative failure cases, which can be primarily categorized into two failure modes: False Negatives and False Positives.

In False Negative cases, failures primarily stem from complex environmental noise and visual features discrepancies. As illustrated in case (a), dense semantic interference from adjacent signage disperses the model's attention, hindering the precise extraction of target features in crowded street scenes. Furthermore, mismatches between colloquial abbreviations on signboards and formal registered names in databases (case b), as well as the visual ambiguity of homogeneous store types (case c), often lead to failed alignments. These examples indicate that despite the model's robustness, it still struggles when visual cues are drowned out by noise or when there is a significant lexical gap between the visual text and the POI record.

In False Positive cases, the primary cause of error is an over-reliance on coarse-grained category alignment at the expense of fine-grained entity verification. As shown in case (d), when the visual signboard and the textual POI data share identical category keywords (e.g., both containing "Law Firm"), the strong semantic similarity can overshadow the distinct instance-level differences in brand names (e.g., "Fabang" vs. "Daoyan"). Consequently, the model incorrectly predicts a match between two different entities belonging to the same category, highlighting a limitation in distinguishing specific named entities when strong categorical priors are present.

To systematically understand the distribution of model failures, we conducted manual verification and statistical analysis on the failure cases within the test set. The proportional distribution of overall error types is illustrated in Fig. 7(b). The statistics reveal that the four typical errors listed in Fig. 8 account for the vast majority (85.3%). Among them, Name-level Semantic Similarity (32.5%) and Dense Semantic Interference (22.4%) occupy the dominant positions and are the primary causes of model misjudgments. In addition, Homogeneous POI Confusion (18.6%) and Abbreviation-Full Name Mismatch (11.8%) also constitute significant sources of failure. The remaining atypical errors, such as extreme occlusion rendering the text manually unrecognizable or ground-truth annotation biases, are uniformly categorized as Others, accounting for 14.7%.

Overall, these failure cases reflect that SignSpotter still has certain limitations in scenarios with high-density semantic interference, inconsistent textual expressions, and highly homogenized brands. Nevertheless, these situations mostly occur under boundary conditions

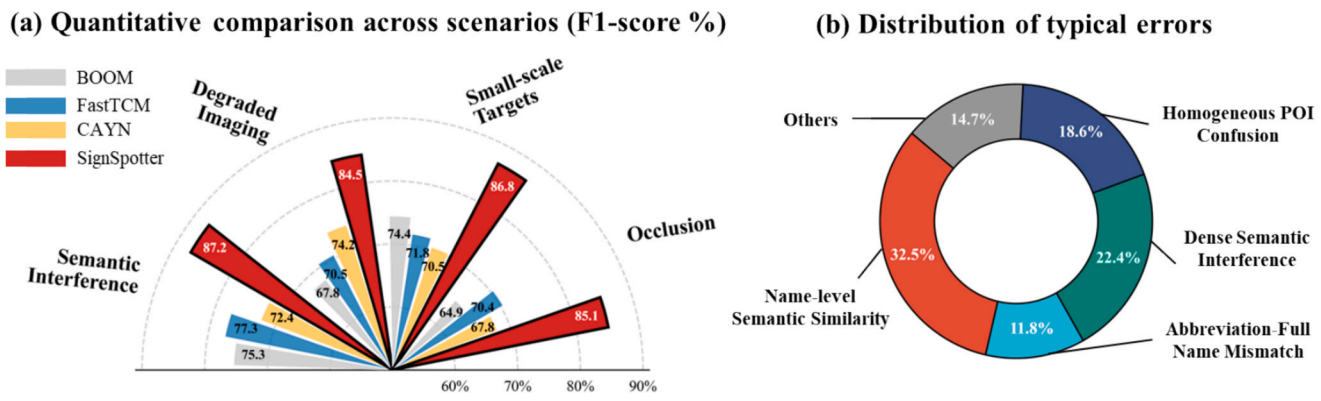


Fig. 7. Quantitative evaluation of SignSpotter under complex street view scenarios.

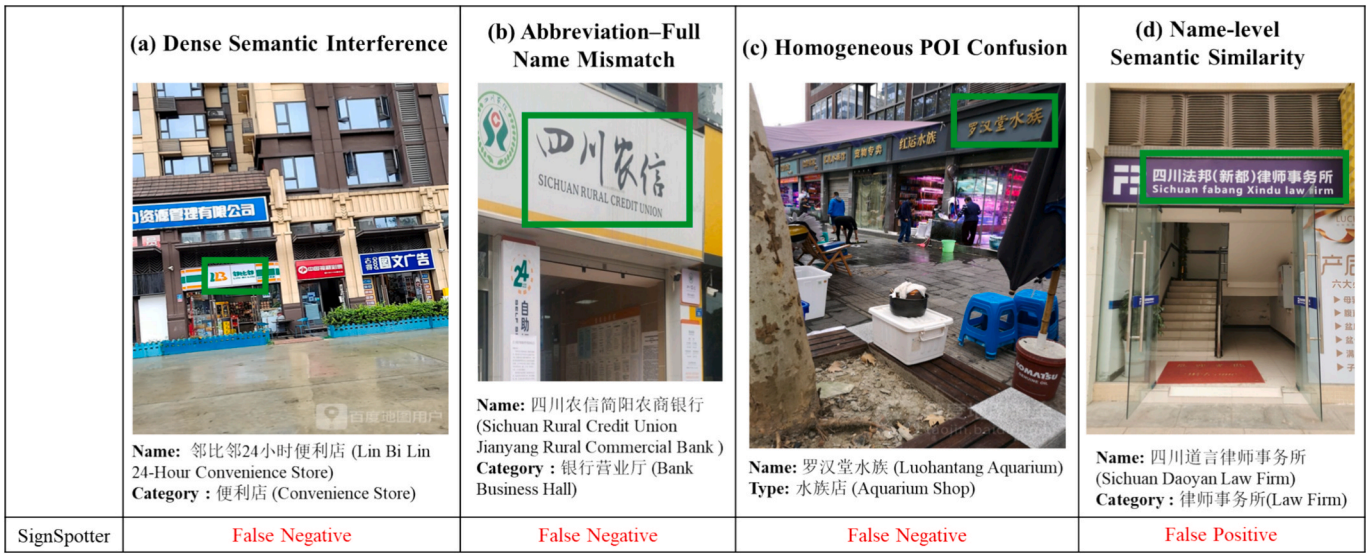


Fig. 8. Representative failure cases of SignSpotter in complex street view scenarios.

where semantic distinction is extremely subtle or visual evidence is highly ambiguous, which does not significantly affect the model's overall performance in conventional street view environments.

4.4. Ablation studies

To further quantify the contribution of key modules within SignSpotter to its overall performance, we conducted systematic ablation experiments on the Chengdu dataset. The ablation design centers on four core components: the Scene Text-Informed Visual Pre-training (STVP) module, the Multi-Scale Visual Tower (MSVT), the Cross-Modal Semantic Reasoning (CMSR) module, and the pre-trained Textual Tower (BERT). By removing or replacing these modules one by one and comparing them with the full model, we evaluated the role of each component in the task of complex street view POI semantic consistency verification. The experimental results are presented in Table 6.

4.4.1. Effect of scene text-informed visual pre-training

Variation A removes the STVP module, retaining only the MSVT and the CMSR module. As observed in Table 6, the F1-score of this variant drops to 79.61%, a decrease of over 11% compared to the full model (90.96%). This indicates that directly employing a general ViT to extract visual features fails to adequately capture the textual semantic information in street view images that is highly relevant to POI attributes.

To further intuitively analyze the mechanism of STVP, we compared and visualized the patch-level feature response distributions of the general ViT and STVP on the same street view images. As shown in Fig. 9, from left to right, the figure displays the original street view image, the patch-level feature activation map of the general ViT and its overlay, and the feature activation map of STVP and its overlay. It can be observed that the high-response regions of the general ViT exhibit a relatively scattered distribution, with attention often covering large background areas while lacking a clear focus on key semantic regions such as storefront text. In contrast, the high-response regions of STVP are distinctly concentrated on signboard text and visual areas highly relevant to POI semantics, while responses from irrelevant background areas are effectively suppressed.

These results demonstrate that the STVP module, through character-level supervision, guides the visual encoder to learn more discriminative text-related visual patterns at the patch level. This significantly enhances the model's perceptual capability regarding textual features in street views and provides a high-quality visual representation foundation for subsequent cross-modal reasoning.

4.4.2. Effect of multi-scale visual tower

Variation B removes the MSVT. In this setting, the model retains STVP and the CMSR module. The F1-score of this variant is 88.98%, showing a certain degree of performance decline compared to the full model. This suggests that multi-scale visual modeling plays a crucial complementary role in enhancing overall performance.

Combining this with the feature response visualization in Fig. 10 allows for a more intuitive understanding of this phenomenon. The first two rows of Fig. 10 illustrate that when multiple text regions or adjacent storefronts coexist in a street view, high-response regions in the single-scale model tend to compete among multiple semantically relevant targets, making it difficult to form stable coverage of the target POI. Although STVP can enhance the overall response intensity of text-related regions, its spatial coverage capability remains limited under a single-scale. In contrast, by introducing multi-scale visual modeling, the model can simultaneously model both local text details and the overall store structure. This makes the feature response more stable in spatial distribution and allows for more accurate focusing on key areas consistent with POI.

From the last two rows of Fig. 10, it can be observed that in complex street view scenarios with multiple adjacent storefronts or small target scales, the single-scale model may only focus on locally prominent areas,

Table 6
Quantitative ablation results evaluating the contribution of key components in SignSpotter.

	Variants	STVP	MSVT	CMSR	BERT	Precision (%)	Recall (%)	F1-score (%)
1	Variation A	×	√	√	√	77.47	81.87	79.61
2	Variation B	√	×	√	√	88.39	89.58	88.98
3	Variation C	√	√	×	√	78.49	86.00	82.07
4	Variation D	√	√	√	×	88.68	87.15	87.91
5	**Full Model**	√	√	√		**90.60**	**91.32**	**90.96**

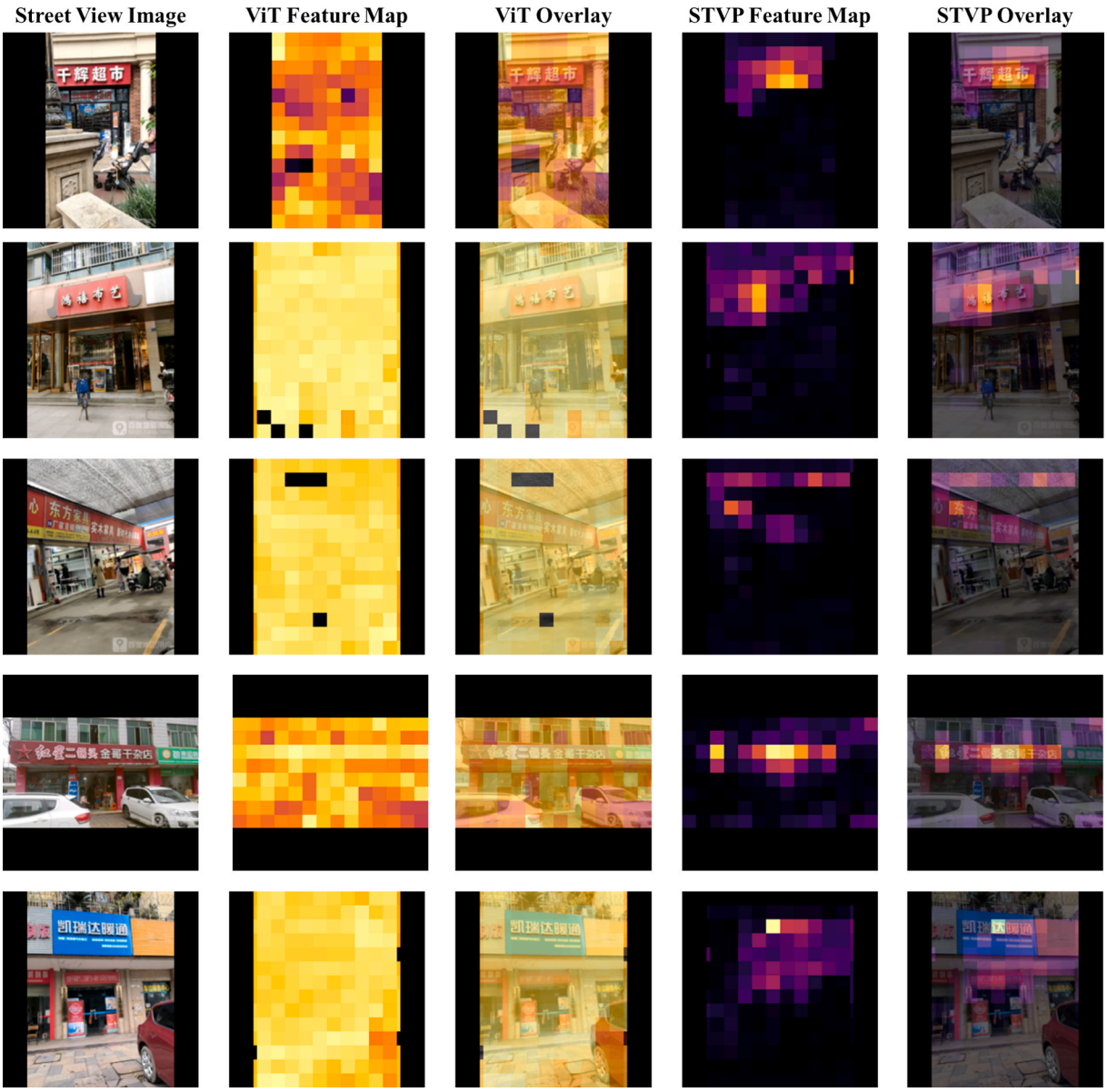


Fig. 9. Visualization of patch-level feature responses extracted by the General ViT and STVP (Warmer colors indicate higher feature activation intensity).

neglecting other potentially relevant semantic targets. Multi-scale modeling, however, can simultaneously activate multiple key areas, significantly improving perceptual capability for small targets and targets with limited spatial distribution, thereby effectively reducing the risk of missed detections. Therefore, the above analysis indicates that the MSVT module plays an indispensable role in enhancing the model's adaptability to street view scenarios characterized by scale heterogeneity.

To deeply analyze the internal multi-scale feature interaction mechanism of MSVT, Fig. 11 further illustrates the response distributions under different scales and feature aggregation strategies. Observing the second column, Fine-Grained Feature Map, it is evident that due to the influence of the 2×2 physical partition boundaries, the signboard text crossing the boundaries is isolated into different local views. This isolation results in spatial discontinuities in the feature responses at the partitions, as indicated by the green dashed boxes in each group of images. The third column, Coarse-Grained Feature Map, is extracted from the complete and unpartitioned original image. Comparing the green dashed boxes at the identical locations reveals that its global receptive field can directly cross the local partition boundaries and completely cover the entire text entity, thereby preserving a coherent physical structure. However, if the coarse and fine features are merely concatenated, as seen in the Simple Concat Feature Map, the model fails to achieve deep alignment of cross scale semantics. As shown in the figure, the high response regions following concatenation exhibit a widespread and divergent distribution, failing to effectively suppress background noise and lacking precise focus on the core signboard text. In contrast, the proposed multi-scale cross attention module, denoted as Cross-Attention Feature Map, can guide local features to perform dynamic addressing and feature interaction within the global context. This mechanism not only effectively stitches the spatial faults caused by the physical grid but also enables the final activation regions to be highly convergent, continuously and precisely focusing on the target text. This objectively verifies the critical role of the cross-attention module in cross scale feature alignment and effective aggregation.

4.4.3. *Effect of cross-modal semantic reasoning*

Variation C removes the CMSR module. In this setting, the model relies on a shallow structure that independently encodes and directly aggregates visual and textual features, lacking explicit cross-modal

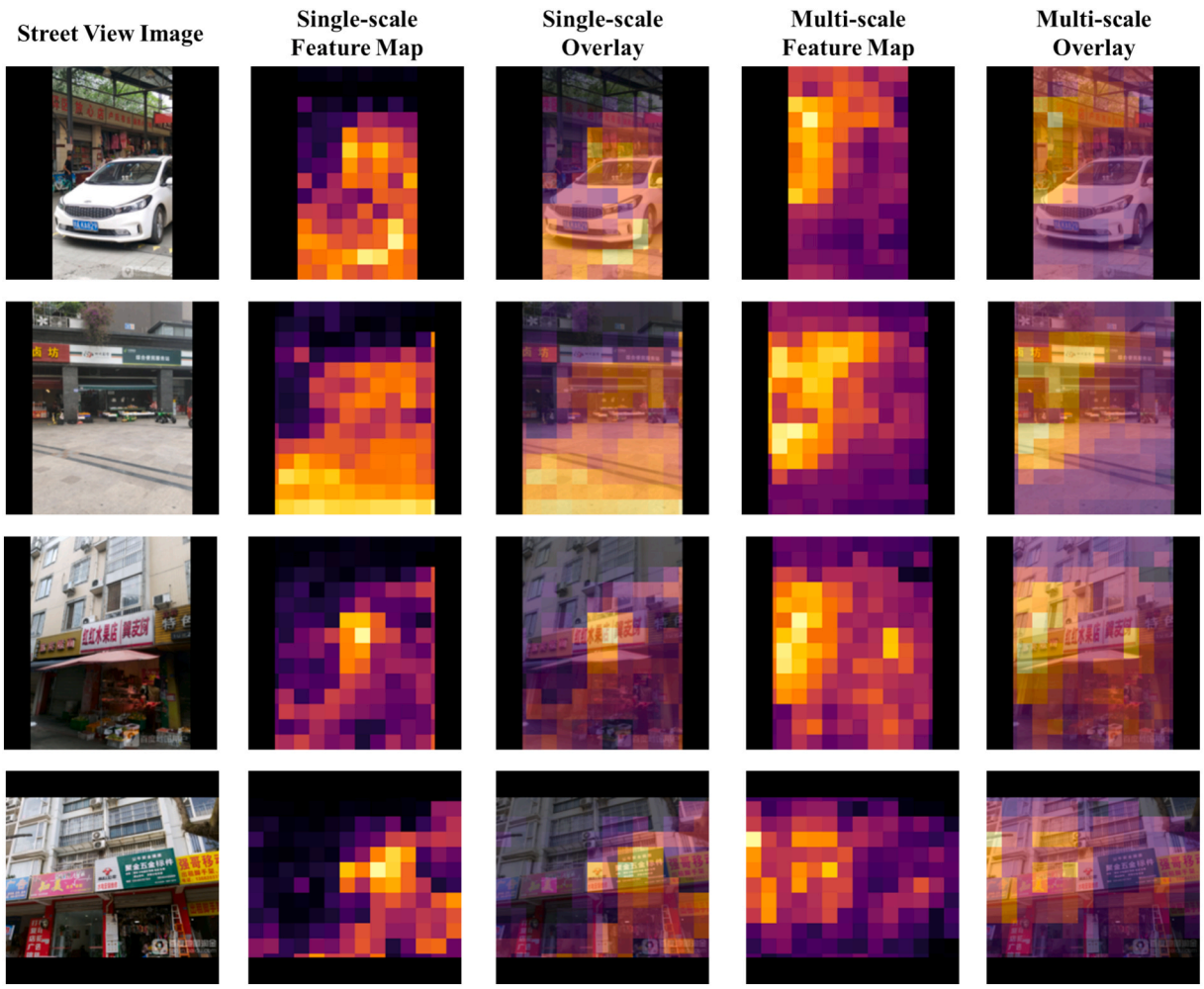


Fig. 10. Visualization of patch-level feature responses extracted by single-scale and multi-scale visual representations (Warmer colors indicate higher feature activation intensity).

semantic reasoning capabilities.

As observed in Table 6, after removing the CMSR module, the model's F1-score drops significantly from 90.96% to 82.07%. This result indicates that even if the visual encoder has learned discriminative representations related to scene text through STVP, the model still struggles to accurately judge the semantic consistency between street view images and POI data in complex environments without a cross-modal reasoning and verification mechanism. Specifically, the CMSR module establishes fine-grained alignment relationships at the visual patch and textual token levels and explicitly models semantic consistency and conflict information between modalities. This enables the model to perform joint reasoning on visual content and POI attributes at the entity-level. Upon removing this module, the discriminative process degenerates into a matching strategy based on coarse-grained feature similarity. In scenarios with complex background interference, semantically highly similar POIs, or coexisting multiple storefronts, this shallow matching approach fails to effectively distinguish true matches from semantically approximate non-target entities, leading to a marked decline in discriminative performance.

To further investigate the contributions of the key operators within the CMSR module, this study conducts a fine-grained ablation study, with the quantitative results presented in Table 7. Variation C1 removes the absolute difference and element wise product operations during the cross-modal feature interaction process. Under this setting, the model only retains the direct concatenation of the aligned visual and textual features. As observed from Table 7, the F1-score of this variant decreases to 88.44%, representing a drop of approximately 2.52% compared to the complete model. This result indicates that the difference and product operations can explicitly quantify the semantic conflicts and consistency between cross-modal features, playing a critical role in the feature interaction process. Lacking these explicit mathematical combination calculations, the model solely relies on the subsequent fully connected layers for non-linear feature mapping, making it difficult to adequately distinguish interfering entities with high visual similarity, thereby leading to a decline in discriminative performance.

Variation C2 degrades the bidirectional cross attention mechanism into unidirectional attention, retaining only the active query of visual features against textual features. As can be observed from Table 7, after removing the bidirectional interaction mechanism, the F1-score of this variant further decreases to 87.81%. This indicates that the bidirectional alignment logic can compel visual features to actively focus on the relevant text, while simultaneously enabling the textual entity to effectively reverse address and verify the corresponding visual clues. Lacking this bidirectional verification mechanism, the model is susceptible to losing critical alignment information when confronting the coexistence of multiple storefronts or background noise interference, subsequently compromising the accuracy of the joint reasoning.

4.4.4. *Effect of pre-trained textual tower*

Variation D removes the pre-trained textual tower (BERT). Under this setting, the model utilizes a randomly initialized trainable word embedding matrix to extract textual features while retaining the STVP,

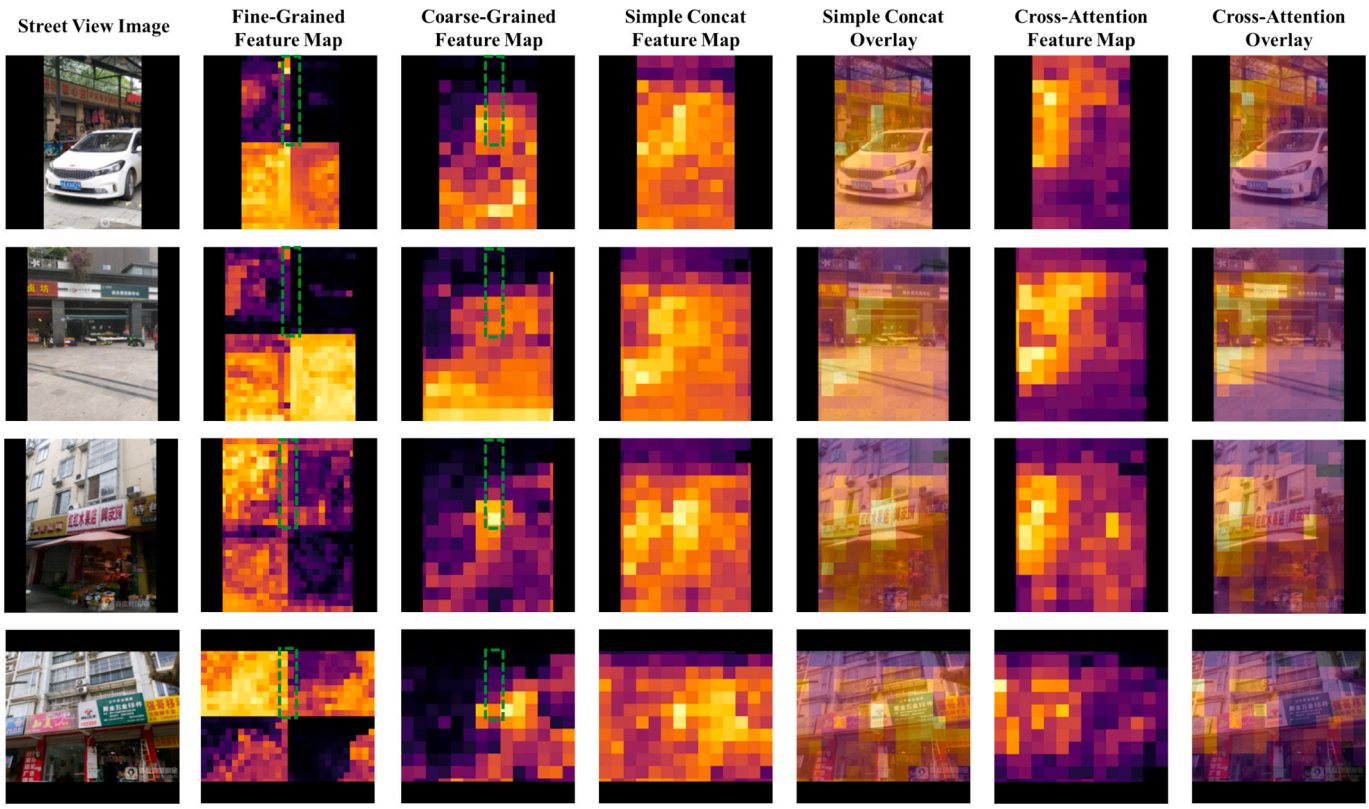


Fig. 11. Visualization of patch-level feature responses across different scales and fusion strategies (Warmer colors indicate higher feature activation intensity).

Table 7
Quantitative results of fine-grained ablation study on the internal mechanisms of the CMSR module.

	Variants	Description	Precision (%)	Recall (%)	F1-score (%)
1	Variation C1	w/o Difference & Product	90.80%	86.21%	88.44%
2	Variation C2	w/o Bidirectional Attention	88.79%	86.84%	87.81%
3	Variation C	w/o CMSR Module	78.49%	86.00%	82.07%

MSVT, and CMSR modules. As can be observed from Table 6, after removing the pre trained language model, the F1 score of this variant significantly drops to 87.91%, representing a decrease of approximately 3.05% compared to the complete model (90.96%). This result indicates that although POI names typically manifest as short texts, learning textual representations from scratch relying solely on the limited supervision signals of the current task makes it difficult to adequately characterize the highly condensed domain information within short texts. In contrast, a pre trained language model adopting the Transformer architecture can leverage its rich prior knowledge to provide highly discriminative textual semantic anchors for visual features. Upon removing this module, the textual encoding capability of the model is restricted, resulting in difficulties in achieving accurate semantic spatial alignment with high quality visual representations in complex street view environments, thereby negatively impacting the final discriminative results.

Synthesizing the above ablation experiments, we can conclude that the STVP, MSVT, CMSR, and the pre-trained Textual Tower in SignSpotter exhibit clear complementarity and synergistic effects. STVP injects scene text semantics into the visual encoder; MSVT enhances the model's adaptability to scale heterogeneity in street views; the pre-trained Textual Tower extracts deep contextualized representations to serve as robust semantic anchors for highly condensed POI names; and CMSR provides the model with entity-level cross-modal semantic reasoning and consistency verification capabilities by effectively aligning these multimodal features. The joint action of these four core components enables SignSpotter to achieve robust and high-precision POI semantic consistency verification in complex real-world street view environments.

4.4.5. *Validation of the multi-scale partition strategy*

To verify the rationality of the proposed 2×2 grid partition strategy in the MSVT module, this study compares the performance of the model under different partition granularities, namely the 1×1 baseline, 2×2, and 3×3 grids. As illustrated by the training and validation curves in Fig. 12, the 2×2 partition strategy achieves the optimal and most stable F1-score. The validation F1-score of the 1×1 baseline is lower than that of the 2×2 strategy. When the partition granularity is further refined to 3×3, the model not only fails to obtain additional accuracy gains but also experiences a substantial performance degradation. Meanwhile, further analysis considering computational complexity reveals that the 3×3 partition strategy increases the number of local feature slices for a single image from 4 to 9. This not only exponentially increases the forward inference time of the visual backbone network but also significantly exacerbates the computational burden of subsequent cross-modal feature interactions. Furthermore, the performance degradation of the model under the 3×3 setting indirectly indicates that an overly dense grid partition increases the risk of truncating the continuous physical structure of scene text, thereby introducing additional boundary noise. Therefore, comprehensively considering prediction accuracy and model computational efficiency, the 2×2 partition achieves the optimal balance between extracting multi-scale visual features and controlling computational overhead.

Additionally, regarding the setting of the input scale, the image resolutions across all branches of the model are fixed at 224×224. This scale strictly aligns with the receptive field and positional encoding of

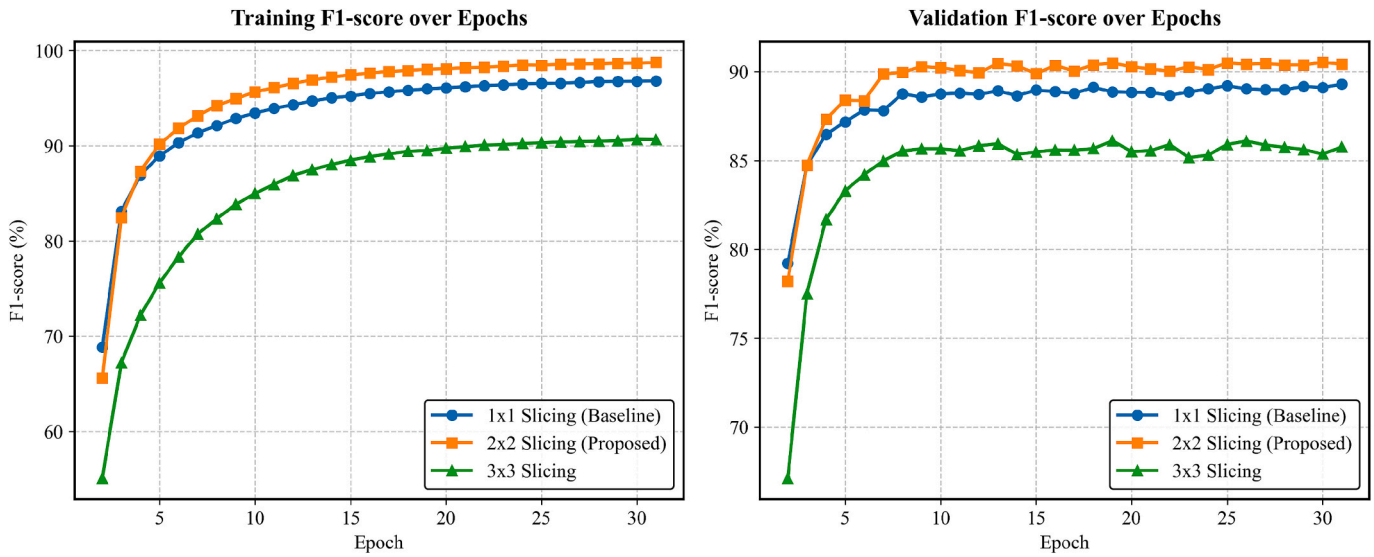


Fig. 12. Comparison of training and validation F1-scores under different multi-scale partition granularities.

the pre trained visual backbone network, ViT B/16. Maintaining the 224 × 224 resolution ensures that each visual region precisely corresponds to a 14 × 14 patch sequence (224 / 16 = 14). This maximizes the transferability of the representations learned during the first stage, namely the STVP, without introducing positional interpolation artifacts.

4.5. Efficiency and complexity analysis

To evaluate the deployment feasibility of the proposed SignSpotter model in real-world large-scale POI consistency verification scenarios, this section conducts a quantitative analysis of its efficiency and computational complexity. Specifically, we introduce three core metrics, the number of model parameters (Params), floating point operations (FLOPs), and average inference latency per image (Latency), to quantify the resource consumption of the model. This section selects the optimal models from each paradigm in Section 4.3.1 as the comparative baselines, primarily including OCR + Edit Distance, ResNet 50 + BERT, BLIP-2, and FastTCM. The comparison results regarding efficiency and accuracy between the baseline models and the proposed method are presented in Table 8.

As shown in Table 8, although the OCR based two stage matching method OCR combined with Edit Distance does not require loading a massive deep learning feature extractor, it heavily relies on the serial decoding process of scene text detection and recognition. This results in an inference latency of 17.81 ms for a single image, and due to the error accumulation of the OCR engine, its F1-score is only 80.31%. The traditional dual tower model ResNet 50 combined with BERT performs best in computational efficiency, with 110.60 M parameters and an inference latency as low as 8.19 ms. However, this model lacks deep cross modal fine grained interaction, which leads to the lowest matching accuracy among all baselines, with an F1-score of only 75.62%. On the other hand, although the generic vision language large model BLIP-2 demonstrates powerful cross modal representation capabilities, its parameter count reaches as high as 3744.76 M, and the single image inference latency surges to 37.18 ms. This exorbitant computational overhead and low-cost effectiveness indicate that directly deploying generic large models struggles to satisfy the actual engineering demands of city scale large scale POI verification.

Table 8
Efficiency and complexity comparison of representative methods.

Methods	Params (M)	FLOPs (G)	Latency (ms/img)	F1 (%)
OCR + Edit Distance	–	–	17.81	80.31
ResNet-50 + BERT	110.60	5.83	8.19	75.62
BLIP-2	3744.76	–	37.18	79.70
FastTCM	128.00	7.30	9.38	85.00
SignSpotter	174.81	86.34	11.92	90.96

In contrast, the proposed SignSpotter model exhibits an excellent trade-off between accuracy and efficiency. Compared with FastTCM, the current most advanced model dedicated to scene text, SignSpotter introduces multi-scale visual feature extraction and fine-grained local cross-modal interaction modules. Although the introduction of this design increases the number of model parameters and computational complexity, benefiting from the end-to-end parallel computing architecture, its single image inference latency is still controlled at 11.92 ms. This is equivalent to efficiently processing 83 verification samples per second. Under the premise of merely adding 2.54 ms of inference time compared to FastTCM, the F1-score of SignSpotter is boosted from 85.00% to 90.96%, achieving an improvement of 5.96%.

In summary, SignSpotter not only effectively overcomes the serial time-consuming bottleneck of traditional OCR but also compensates for the weak interaction capability of dual tower models, while simultaneously avoiding the redundant overhead of large models. While maintaining high inference efficiency, it achieves the current best matching accuracy, fully possessing the feasibility for practical deployment in large scale real world urban scenarios.

5. Conclusions

This study demonstrates that integrating scene text priors is essential for parsing complex urban environments. By proposing the SignSpotter framework, we showed that explicit scene text-informed pre-training and multi-scale cross-modal alignment effectively overcome the inherent challenges of scale variation and background noise in verification tasks. Quantitative evaluations confirm the robustness of the proposed method, achieving an optimal F1-score of 90.96% on the source dataset and maintaining stable generalization with F1-scores exceeding 86% in cross-city and cross-platform transfer tests. Ultimately, this framework provides a scalable, cost-effective paradigm for mapping agencies to automate the maintenance of databases, facilitating the transition toward efficient human-in-the-loop updating pipelines.

Despite these advancements, limitations remain. The current model exhibits performance drops in scenarios with extreme semantic interference, and the heavy architecture of the model may hinder real-time edge deployment. Furthermore, while the logic of cross-modal

consistency is inherently region-agnostic, applying the model directly to foreign cities poses challenges related to language differences and architectural domain shifts. Future work will focus on integrating multilingual OCR engines, exploring lightweight architectures, and incorporating auxiliary geospatial constraints to enhance the global applicability of the framework and real-time performance.

CRediT authorship contribution statement

Pengpeng Li: Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Biao Pei:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Yuting Zhang:** Writing – review & editing, Visualization, Validation, Funding acquisition, Data curation. **Qing Zhu:** Writing – review & editing, Supervision, Conceptualization. **Tao Liu:** Writing – review & editing, Validation, Investigation, Data curation. **Jiping Liu:** Writing – review & editing, Resources, Investigation. **Ping Du:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Detailed Mathematical Formulations

This appendix provides the standard mathematical definitions omitted from the main text for readability.

A.1. Optimization objective for scene text-informed visual pre-training

In the pre-training stage (Section 3.2.2), the CTC is formulated to maximize the conditional probability of the target character sequence Y given the input image X:

$$\mathscr{L}_{CTC} = -\log \sum_{\pi \in \mathscr{B}^{-1}(Y)} P(\pi|X) \tag{A.1}$$

where π denotes an alignment path that includes the blank symbol, and $\mathscr{B}(\cdot)$ represents the mapping operation that removes repeated characters and blank labels.

A.2. Resolution-adaptive multi-scale visual feature extraction

In the Multi-Scale Visual Tower (Section 3.3.1), different operations are applied based on the input resolution. For the High-Resolution Strategy ($H, W \geqslant 224$), the fine-grained and coarse-grained feature sequences (v_{fine} and v_{coarse}) are formulated as:

$$v_{fine} = \{f_{enc}(R(p_i))\}_{i=1}^{4} \tag{A.2}$$

$$v_{coarse} = f_{enc}(R(I)) \tag{A.3}$$

where $R(\cdot)$ denotes the resizing operation to the standard resolution (224×224), p_i represents the i-th crop of a 2×2 grid, and $f_{enc}(\cdot)$ is the visual encoder.

For the Low-Resolution Strategy ($H, W < 224$), to maintain structural consistency, the corresponding feature sequences are defined as:

$$v_{fine} = \{f_{enc}(R(I))\} \cup \{0_{pad}\}_{j=1}^{3} \tag{A.4}$$

$$v_{coarse} = f_{enc}(R(\uparrow(\downarrow_{\times 2}(I)))) \tag{A.5}$$

where 0_{pad} represents zero tensors used to align dimensions, and $\downarrow_{\times 2}$ and $\uparrow$ denote down-sampling by a factor of 2 and subsequent up-sampling, respectively.

A.3. Standard multi-head attention mechanism

In the Cross-Modal Alignment module (Section 3.4.1), the standard multi-head attention mechanism $MultiHead(\cdot)$ is used to achieve soft alignment between visual patches and textual tokens, which is defined as follows:

$$MultiHead(Q, K, V) = Concat(head_1, \ldots, head_n) \tag{A.6}$$

$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^v) \tag{A.7}$$

where $head_i$ denotes the output of the i-th attention head, n denotes the number of attention heads, and $Concat(\cdot)$ denotes the feature concatenation operation. The function $Attention(\cdot)$ denotes to the scaled dot product attention mechanism.

4.4. Semantic composition, pooling, and consistency classifier

In the Discriminative Reasoning and Consistency Inference module (Section 3.4.2), the enhanced visual and textual feature sequences are fed into multi-layer Transformer encoders for semantic composition:

$$V_{composed} = TransformerEncoder(V_{enhanced}) \tag{A.8}$$

$$T_{composed} = TransformerEncoder(T_{enhanced}) \tag{A.9}$$

Subsequently, a hybrid pooling strategy is adopted to map variable-length feature sequences into fixed-length global representations:

$$V_{pooled} = Concat\big(MaxPool(V_{composed}), AvgPool(V_{composed})\big) \tag{A.10}$$

$$T_{pooled} = Concat\big(MaxPool(T_{composed}), AvgPool(T_{composed})\big) \tag{A.11}$$

Max pooling is used to capture the most salient discriminative cues, while average pooling preserves information about the overall semantic distribution.

Finally, the global visual and textual representations are concatenated and fed into a fully connected network for semantic consistency prediction:

$$h = W_1[V_{pooled}; T_{pooled}] + b_1 \tag{A.12}$$

$$\hat{y} = \mathrm{Softmax}(W_2 h + b_2) \tag{A.13}$$

where $\hat{y}$ denotes the predicted probability that the street view image and the POI data are semantically consistent. W_1 and W_2 are learnable weight matrices; b_1 and b_2 are bias terms.

Data availability

Data will be made available on request.

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